National 5 Analogue Electronics Past Paper Questions

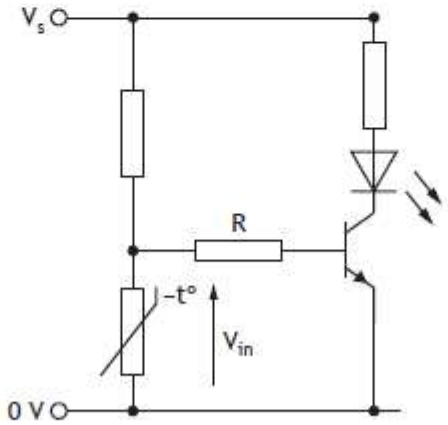
Q1 (2025, Q14)

14. A carpet cleaner is shown below.



The carpet cleaner uses a heater to maintain the temperature of the water in a clean water tank.

A simplified circuit, used to indicate when the water is too cold, is shown below.



(a) Describe the operation of the circuit as the water temperature increases. Include reference to the resistance of the thermistor and the voltage V_{in}. 4

As the water temperature increases . . . _____

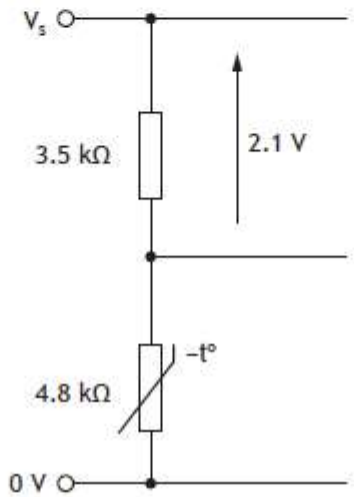
… the thermistor's resistance will decrease (Temperature Up, Resistance Down – TURD). This will cause V_{in} to reduce. When V_{in} drops to below ~0.7V, the transistor will switch off meaning no current flows between the transistor's collector and emitter which will result in the LED switching off.

14. (continued)

(b) Explain the purpose of resistor R in the circuit shown opposite. 2

The resistor will limit current flowing to the transistor base and will therefore protect the transistor.

The sensing part of the circuit is shown below.



(c) Calculate the supply voltage V_s in the circuit shown above. 4

There are two methods:

Method 1:

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$V_2 = \frac{R_2}{R_1} \times V_1$$

$$V_2 = \frac{4800}{3500} \times 2.1 = 2.88V$$

$V_{supply} = V_1 + V_2$

$V_{supply} = 2.1 + 2.88 = 4.98 =$ **5.0V**

Method 2:

V = IR

I = V / R = 2.1 / 3500 = 0.0006A

Current in series circuits is the same everywhere, and so;

$V_2 = IR_2 = 0.0006 \times 4800 = 2.88V$

$V_{supply} = V_1 + V_2$

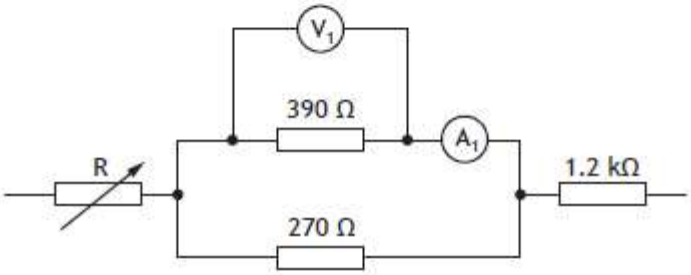
$V_{supply} = 2.1 + 2.88 = 4.98 =$ **5.0V**

Q2 (2024, Q8)

8. A child's microphone with built-in speaker is shown.



Part of the circuit used to control the volume of the speaker is shown.



(a) For the circuit shown above:

(i) calculate the resistance of the parallel branch 2

Two methods:

Method 1:

$$R_{parallel} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{390 \times 270}{390 + 270} = 159.545454$$

R_T = **160Ω** to 2 sig. fig

Method 2:

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{parallel}} = \frac{1}{390} + \frac{1}{270} = 0.00256 + 0.00370 = 0.00627$$

$R_{parallel}$ = 1 / 0.00627 = 159.545454 = **160Ω** to 2 sig. fig

8. (a) (continued)

(ii) calculate the total resistance of the circuit when the variable resistor R is set to 780 Ω. 2

R_{total} = 780 + 160 + 1200 = 2140 = **2100Ω = 2.1kΩ** to 2 sig. fig

(b) Calculate the current reading on ammeter A_1 when the voltage V_1 is 1.2 V. 3

V = IR

I = V / R = 1.2 / 390 = 0.003076923 = **0.0031A** =**3.1mA** to 2 sig. fig.

The resistance of the variable resistor R in the circuit is increased.

(c) State the effect this will have on the:

(i) voltage V_1 will decrease (voltage across variable resistor will increase but supply voltage remains constant and so voltages across other components must reduce) 1

(ii) current A_1 will decrease (overall resistance increase and so current must reduce) 1

During the development of the child's microphone and speaker, computer simulation was used to test the complete circuit before constructing a prototype.

(d) Explain why it is an advantage to test the circuit using computer simulation before constructing the prototype. 2

Components are not destroyed ...making it cheaper to test.

A range of components are readily available ...making it quicker to change/test.

Software can highlight issues in the circuit ...making it easier to correct faults.

It is safer to the user ...because fault/failure is simulated.

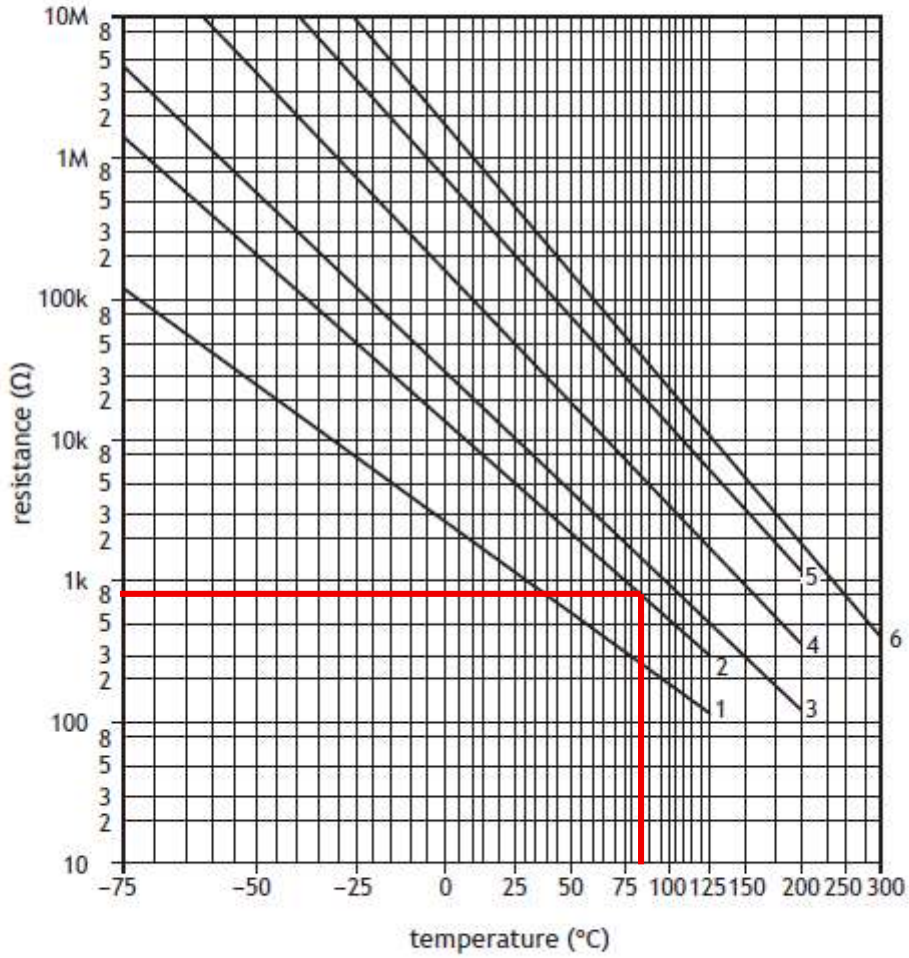
Q3 (2024, Q10)

10. A kettle is shown below. MARK



A thermistor is used as part of a sensor to measure the water temperature in the kettle.

The operating characteristics for a range of thermistors are shown on the graph below.

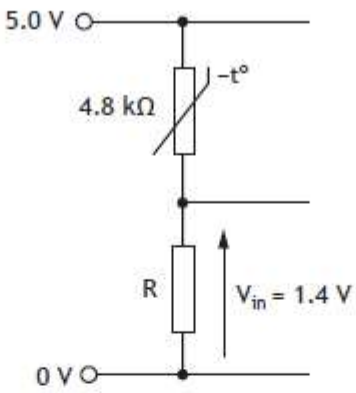


(a) Determine the resistance of a type 2 thermistor at a temperature of 85 °C, with reference to the graph. 1

800Ω

10. (continued)

The sensing circuit used to detect the temperature of the water is shown below.



(b) State the name of this type of series circuit. 1

Voltage Divider (or Potential Divider

(c) Calculate the resistance R in the sensing circuit shown above. 4

$V_{supply} = V_1 + V_2$

$V_1 = V_{supply} - V_2 = 5 - 1.4 =$ **3.6V**

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

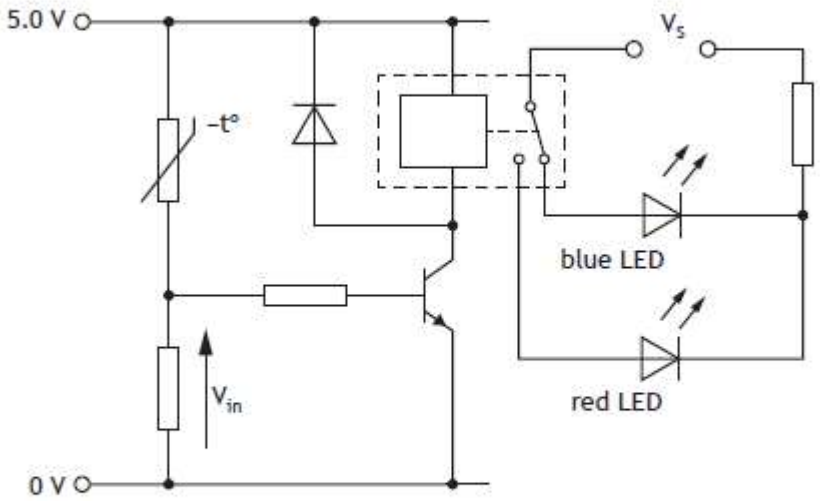
$$R_2 = \frac{V_2}{V_1} \times R_1$$

$$R_2 = \frac{1.4}{3.6} \times 4800 = 1866.6666667 = \mathbf{1.9k\Omega}$$

10. (continued)

The kettle has coloured LEDs to indicate the temperature of the water.

A simplified circuit used to model the operation of the LEDs is shown below.



(d) Describe the operation of this circuit as the water temperature increases.

Include reference to the resistance of the thermistor, the voltage V_{in} and both LEDs. 5

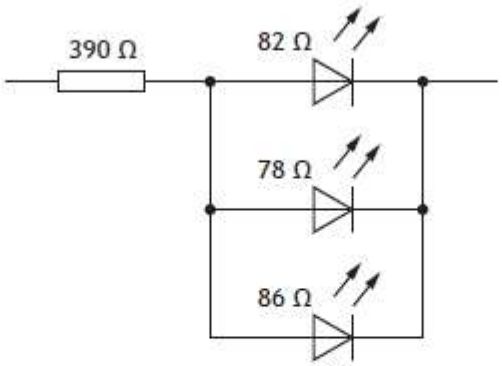
As the water temperature increases . . . _____

… the thermistor's resistance will decrease (Temperature Up, Resistance Down – TURD). This will cause V_{in} to increase. When V_{in} increases to ~0.7V, the transistor will switch on meaning current flows between the transistor's collector and emitter which will result in the relay activating. This will result in the blue LED switching off and the red LED switching on.

Q4 (2023, Q10d)

10. (continued)

Part of the circuit used in the warning sign is shown below.



(d) (i) Calculate the resistance of the three LEDs in parallel. 3

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{parallel}} = \frac{1}{82} + \frac{1}{78} + \frac{1}{86} = 0.012195 + 0.012821 + 0.011628$$

$$= 0.0366435$$

$R_{parallel}$ = 1 / 0.0366435 = 27.289938 = **27Ω** to 2 sig fig

(ii) Calculate the total resistance of the circuit shown above. 1

R_{total} = 390 + 27 = 417 **= 420Ω** to 2 sig. fig

10. (continued)

The warning sign is rated at 12 V, 1.8 W.

(e) Calculate the current supplied to the warning sign. 3

P = IV

I = P / V = 1.8 / 12 = **0.15A**

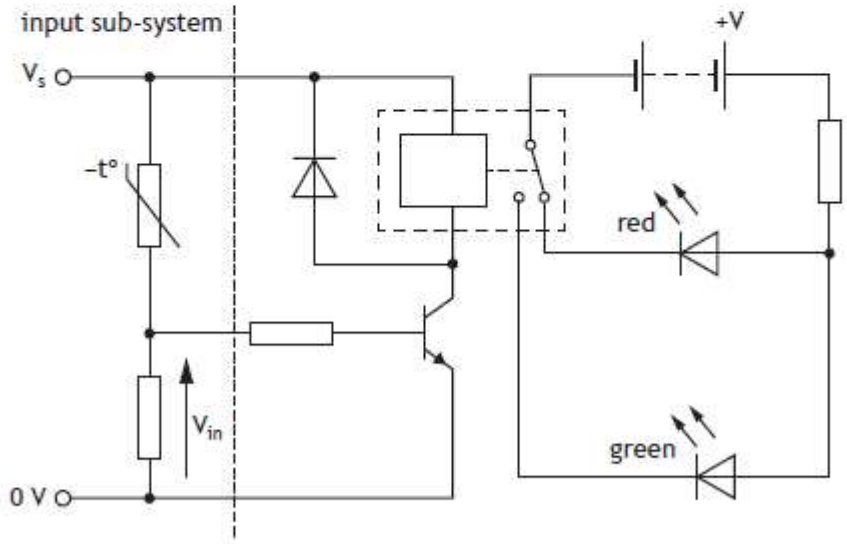
Q5 (2023, Q14)

14. A laminator is used to cover a sheet of paper in a plastic sleeve.



The plastic sleeve is heated by the laminator as it passes through.

The circuit used to indicate when the laminator is at the correct temperature is shown below.



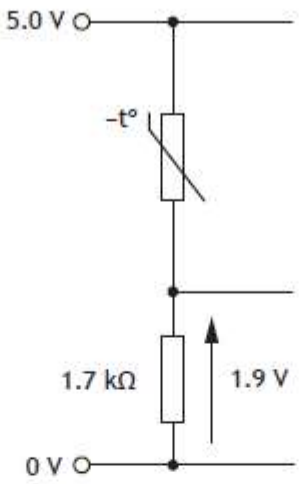
(a) Describe the operation of the input sub-system as the temperature increases. Make reference to the resistance of the thermistor and the voltage V_{in}. 2

As the temperature increases . . .

… the thermistor's resistance will decrease (Temperature Up, Resistance Down – TURD). This will cause V_{in} to increase. When V_{in} increases to ~0.7V, the transistor will switch on meaning current flows between the transistor's collector and emitter which will result in the relay activating. This will result in the red LED switching off and the green LED switching on.

14. (continued)

The input sub-system is shown below.



(b) Calculate the resistance of the thermistor for the condition shown above. 4

There are two methods.

Method 1:

$V_{supply} = V_1 + V_2$

$V_1 = V_{supply} - V_2 = 5 - 1.9 = 3.1V$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$R_1 = \frac{V_1}{V_2} \times R_2$$

$$R_1 = \frac{3.1}{1.9} \times 1700 = 2773.6842$$

$$= \mathbf{2.8k\Omega} \text{ to 2 sig. fig.}$$

Method 2:

$V = IR$

$I = V / R = 1.9 / 1700 = 0.00111765A$

$R_{thermistor} = V / I$

$R_{thermistor} = 3.1 / 0.00111765$

$R_{thermistor} = 2773.684211 =$ **2.8kΩ** to 2 sig.fig.

(c) Describe, with reference to the circuit shown opposite, the effect on the red LED and the green LED when the transistor first activates the relay. 2

When the transistor activates the relay . . .

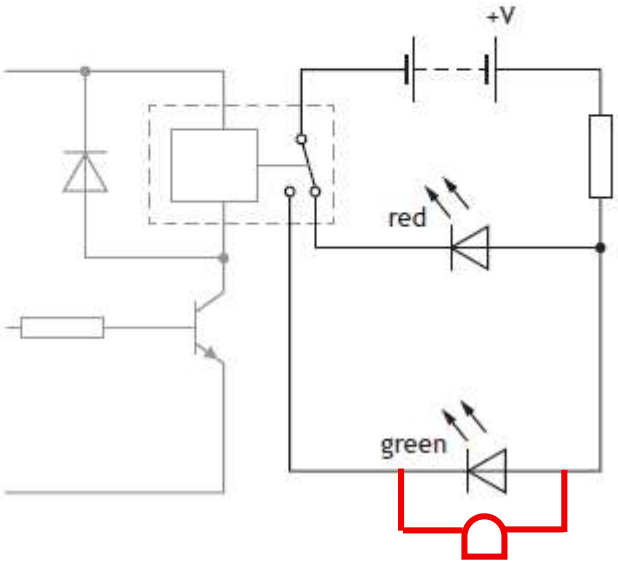
The red LED will switch off.

The green LED will switch on.

14. (continued)

A possible modification to the circuit is to include an audible alert when the laminator is at the correct temperature.

(d) Draw, in the circuit below, the symbol for a buzzer connected in parallel with the green LED. 2



An electronic engineer used computer simulation to test the modified circuit.

(e) Describe an advantage of using computer simulation compared to building a circuit for testing purposes. 1

Quicker to develop solution / Saves money as no replacement components needed / Components will not be destroyed or wasted / Easier to change circuit during testing / No risk to the user due to fault.

A large reduction in speed is required for a feed roller to slowly move the plastic sleeve and paper through the laminator.

(f) Explain why a compound gear train is more suitable than a simple gear train to create this large reduction in speed in the laminator. 2

A compound gear train will use smaller gears... which makes it able to fit inside the small space/laminator.